

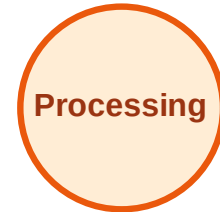
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



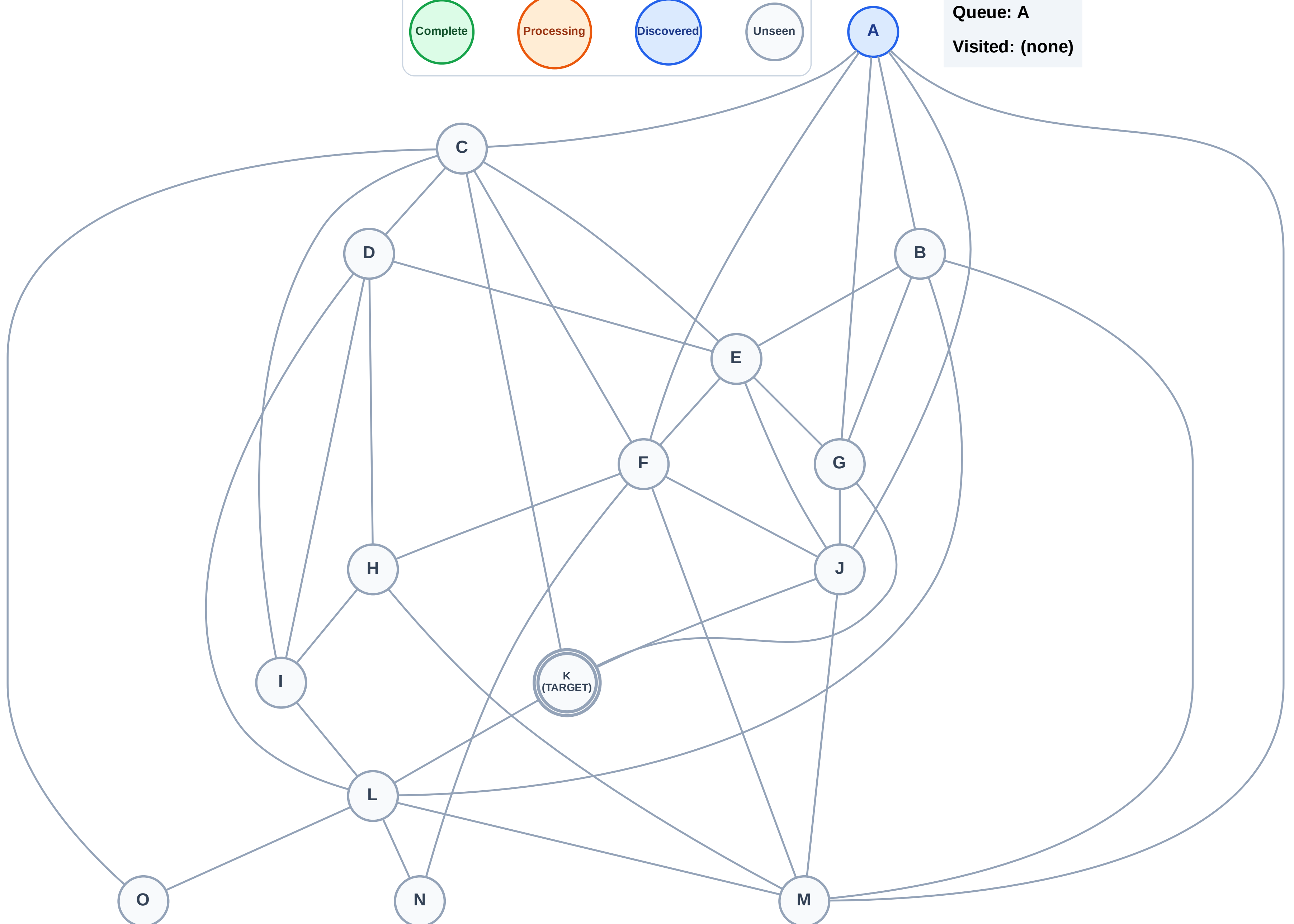
Discovered



Unseen

Queue: A

Visited: (none)



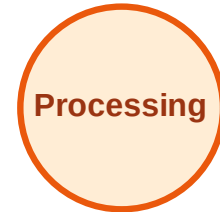
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



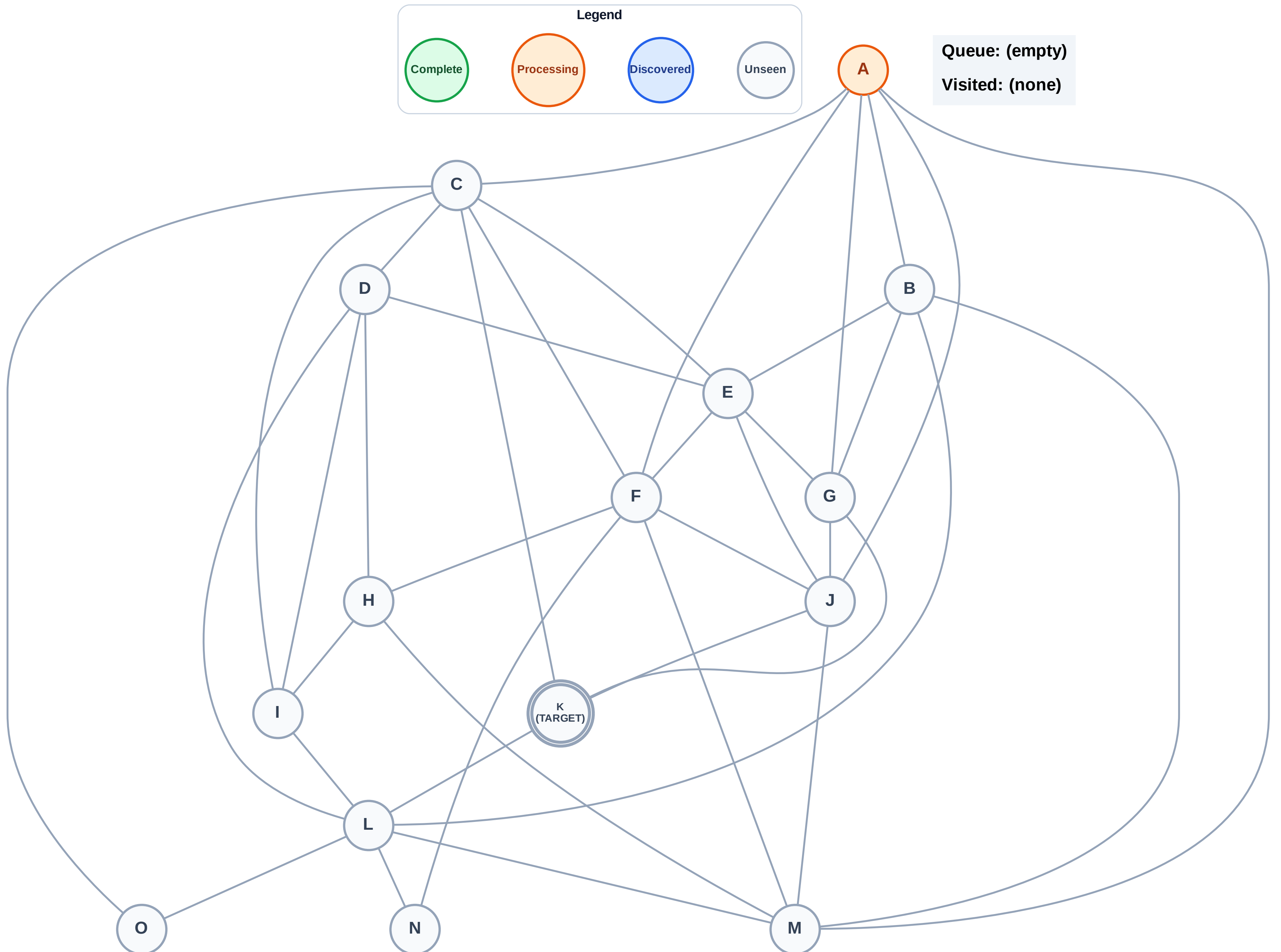
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

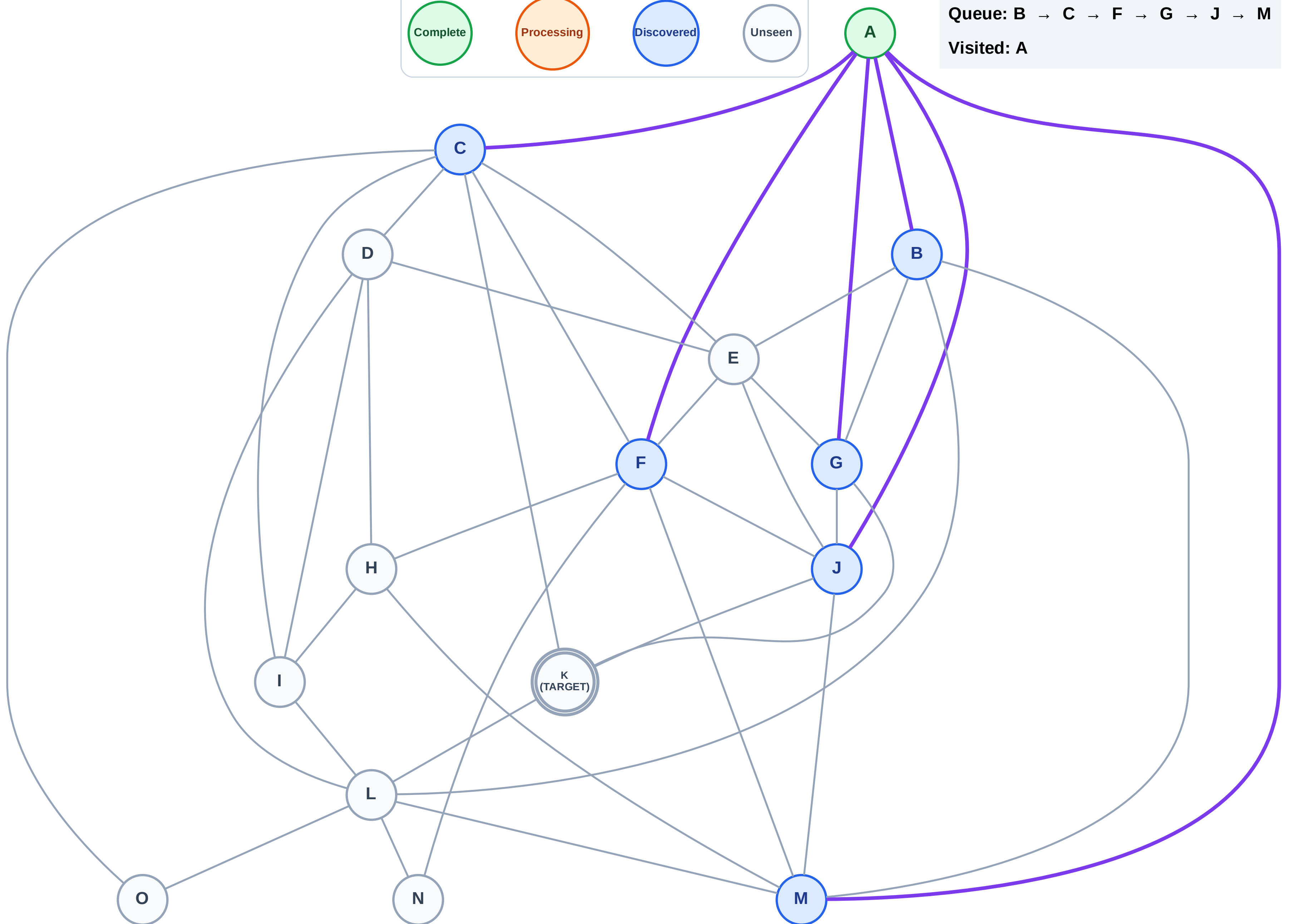
Step 3: Discover B, C, F, G, J, M from A

Legend



Queue: B → C → F → G → J → M

Visited: A



BFS Traversal

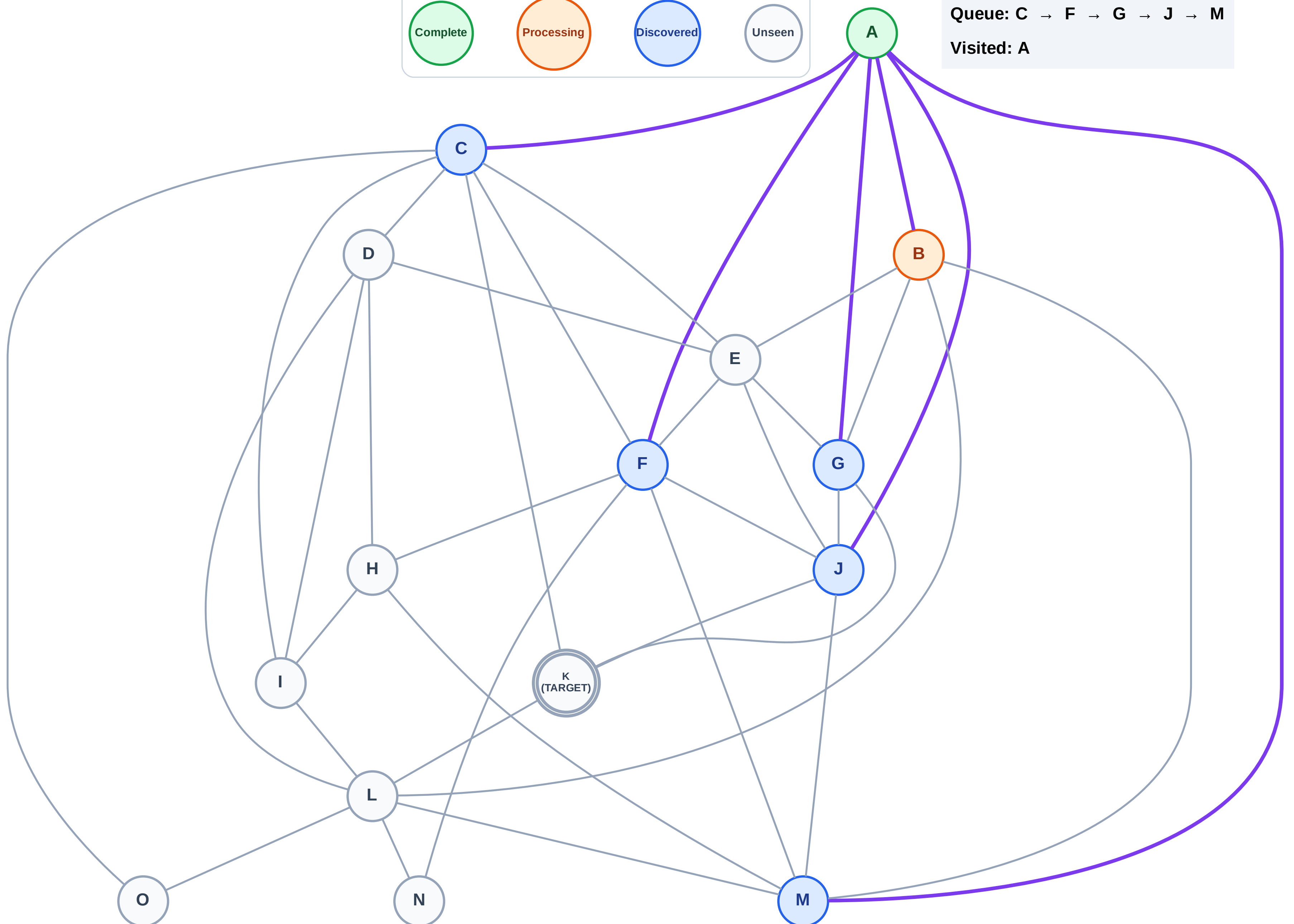
Step 4: Process node B

Legend



Queue: C → F → G → J → M

Visited: A



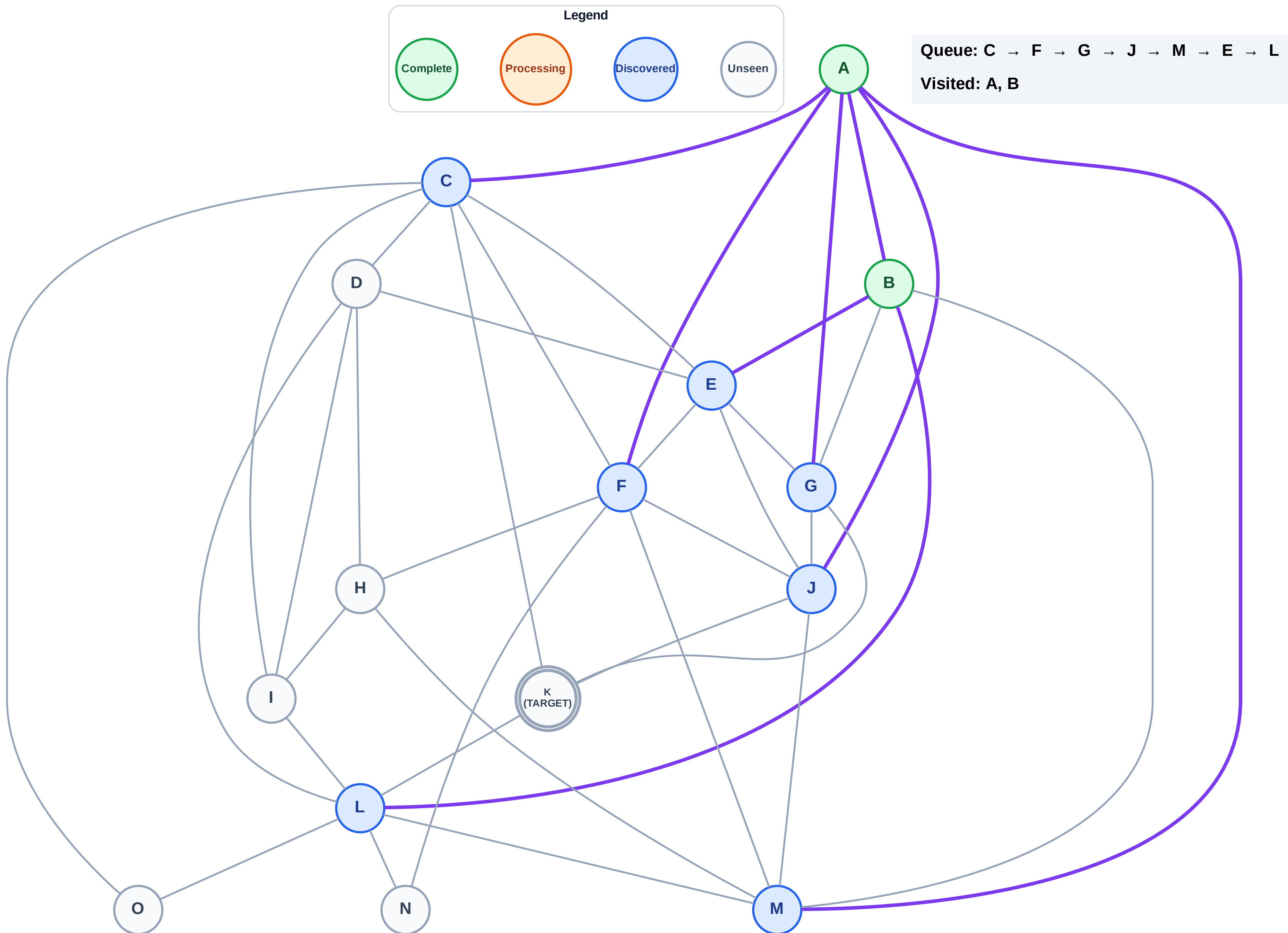
BFS Traversal

Step 5: Discover E, L from B

Legend

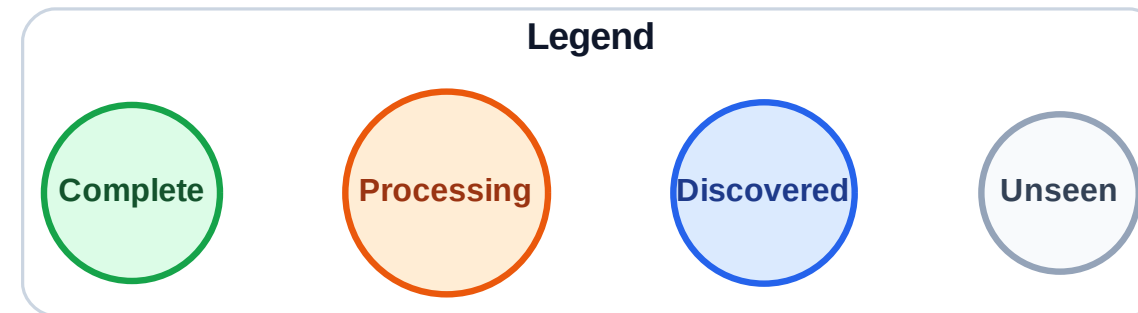


Queue: C → F → G → J → M → E → L
Visited: A, B

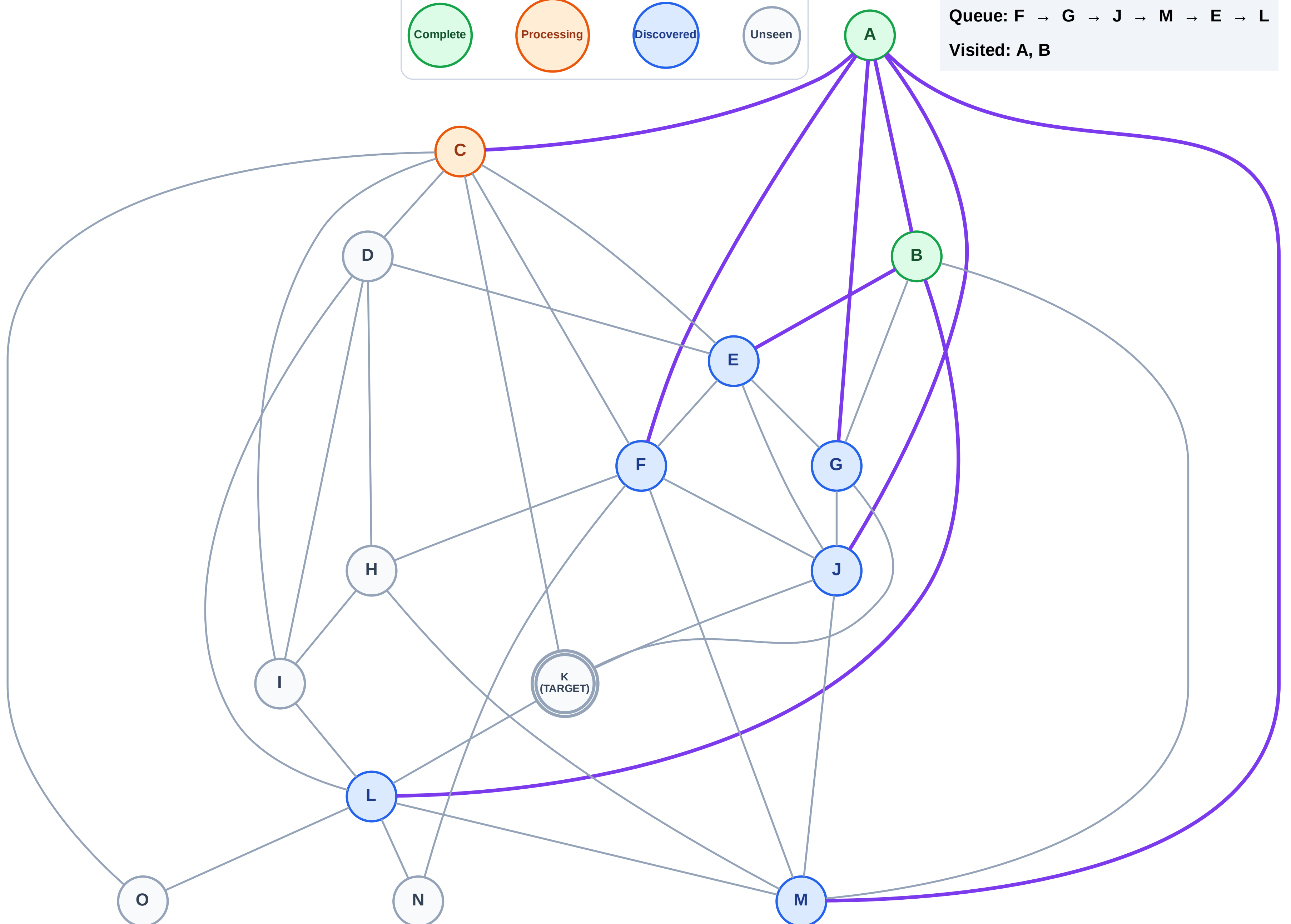


BFS Traversal

Step 6: Process node C



Queue: F → G → J → M → E → L
Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

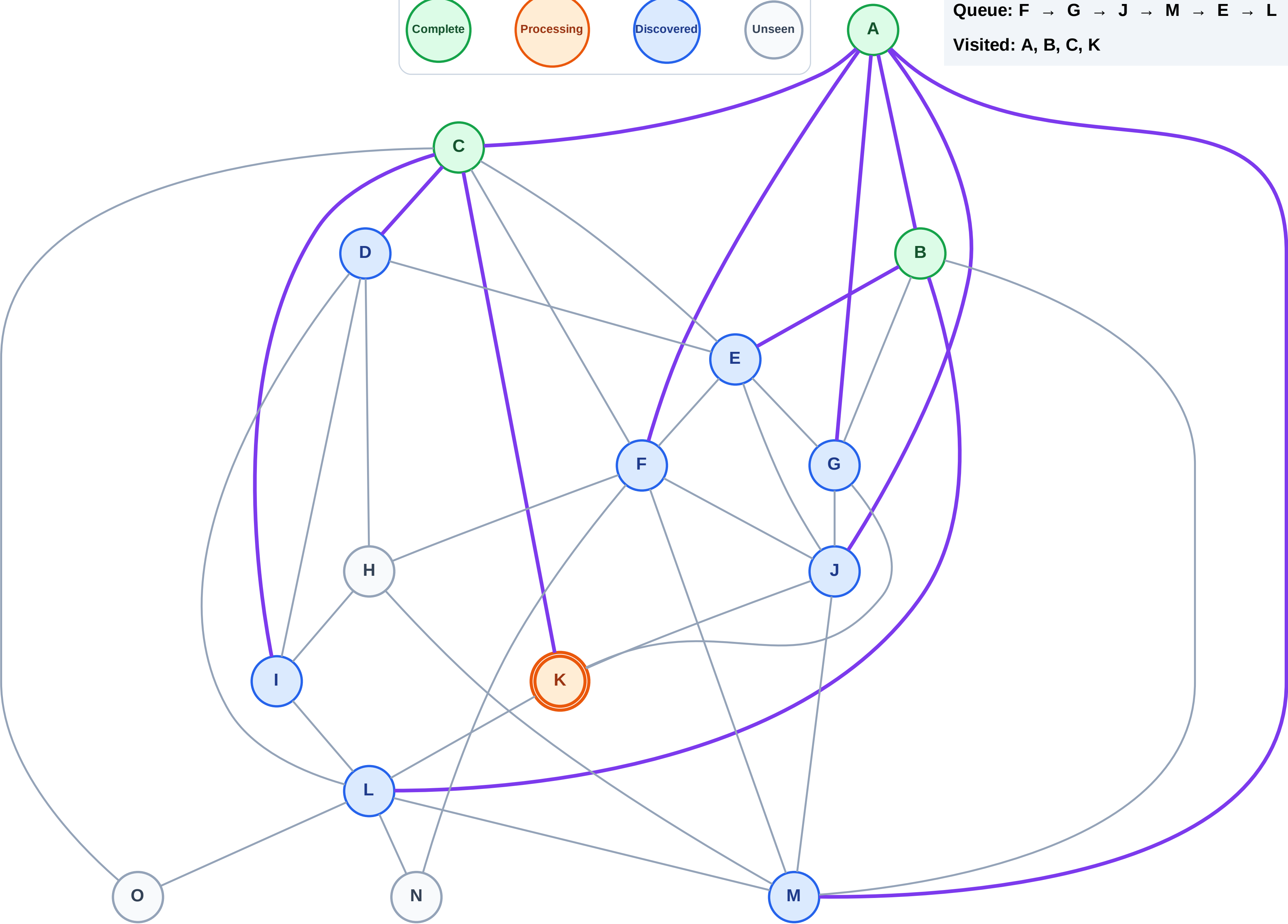
Processing

Discovered

Unseen

Queue: F → G → J → M → E → L → D → I

Visited: A, B, C, K



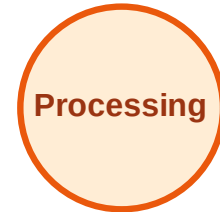
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

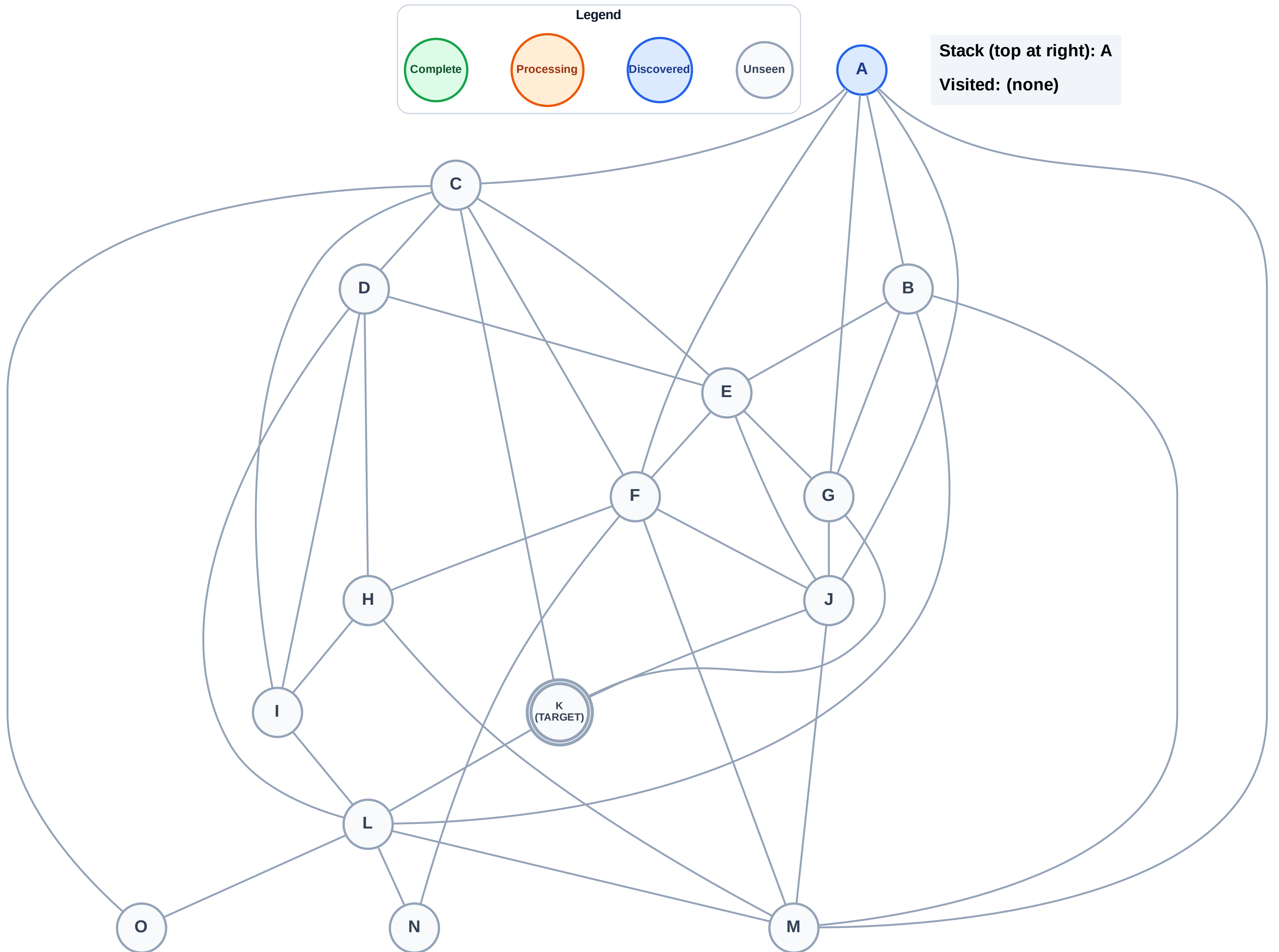


Discovered



Unseen

Stack (top at right): A
Visited: (none)



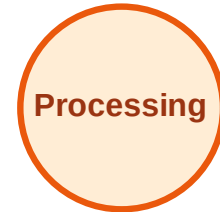
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



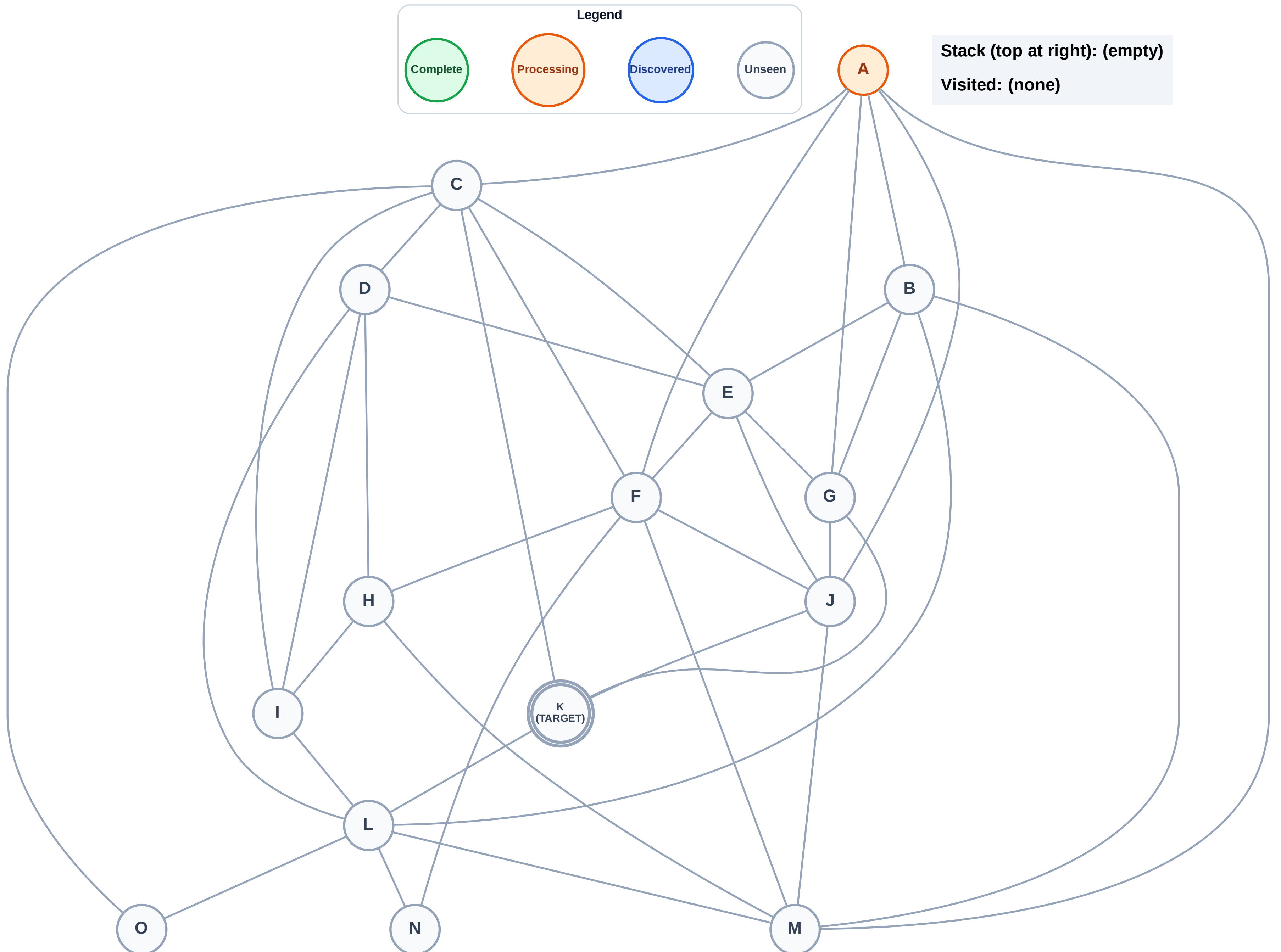
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

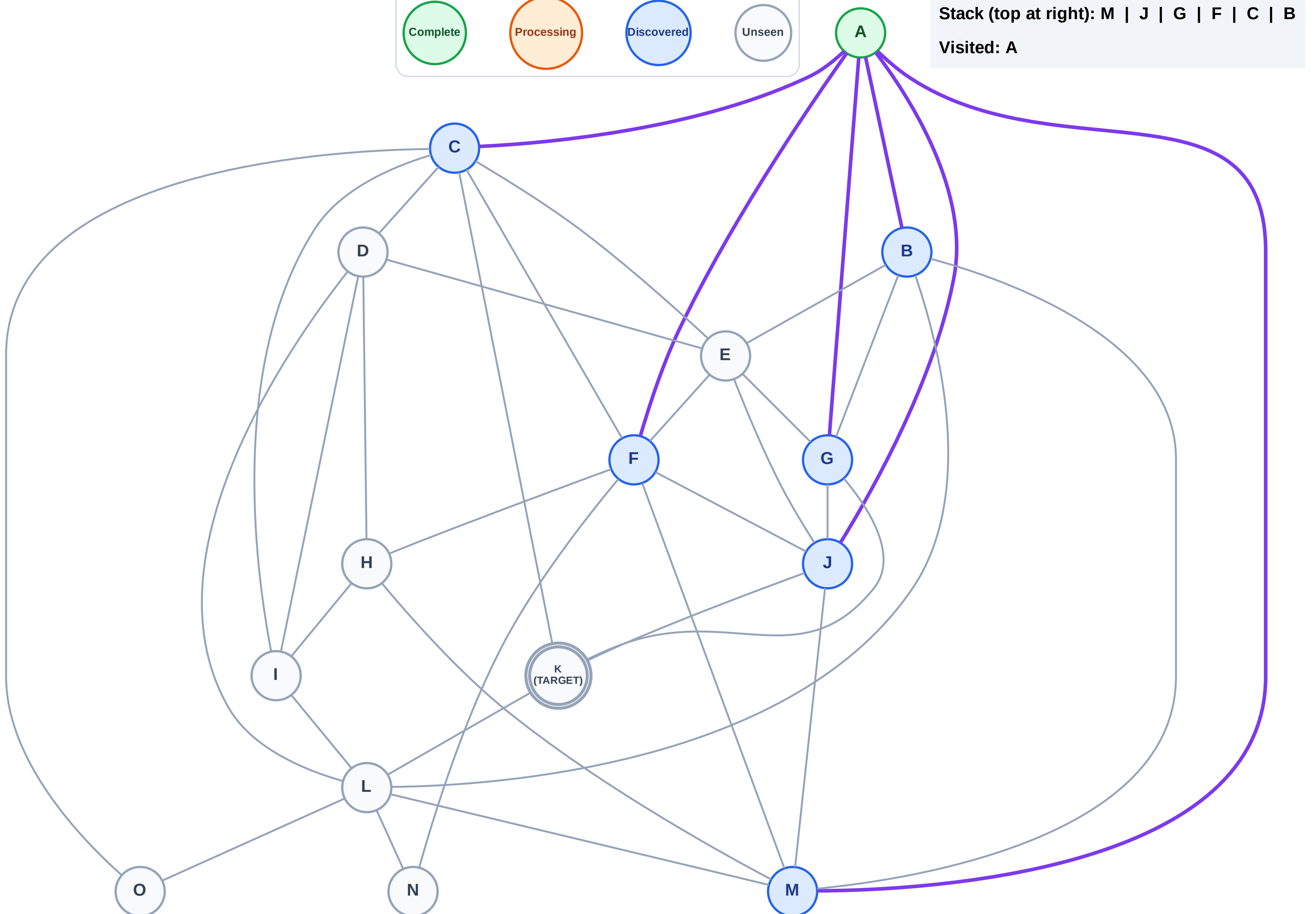
Step 3: Discover M, J, G, F, C, B from A

Legend



Stack (top at right): M | J | G | F | C | B

Visited: A



DFS Traversal

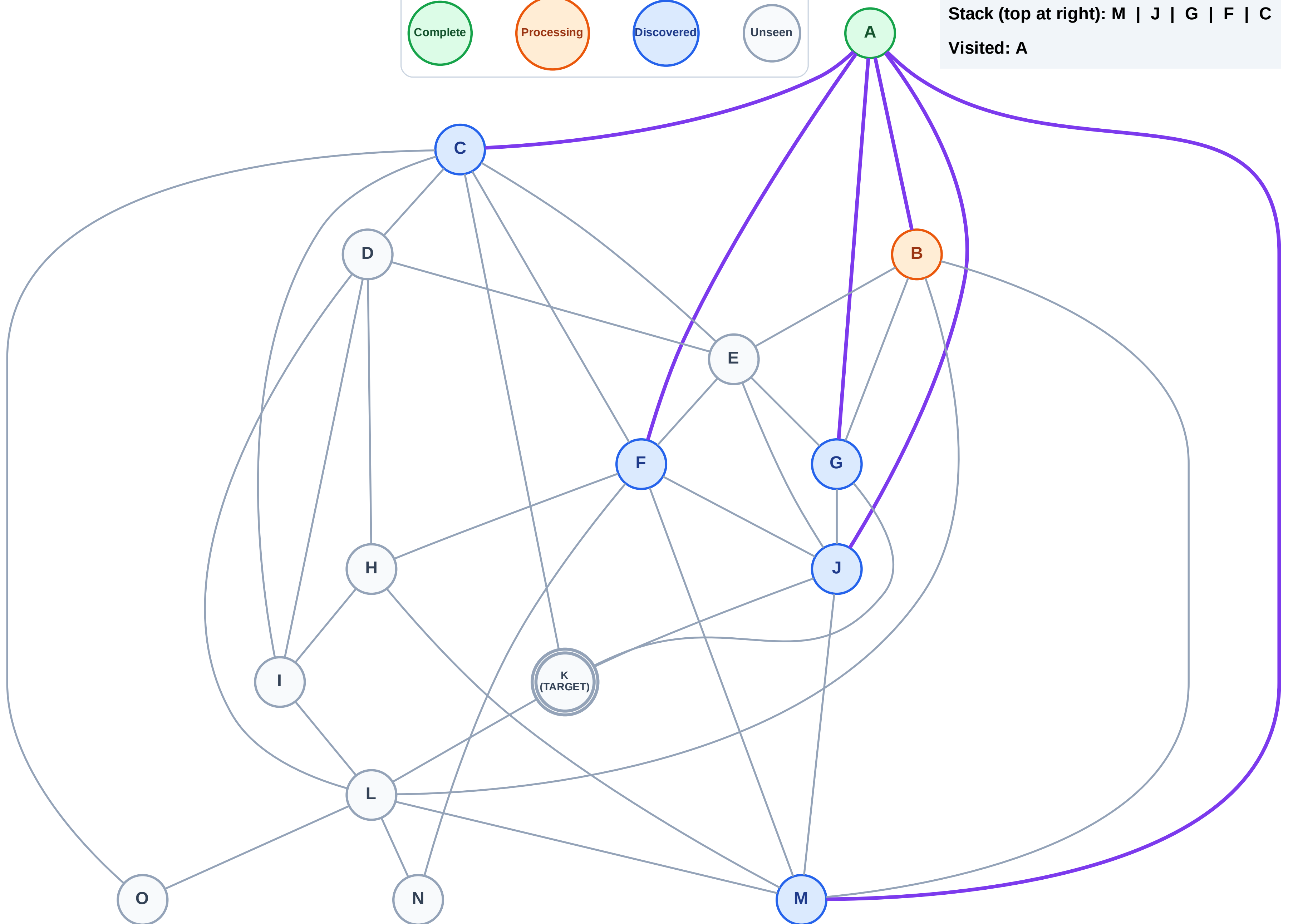
Step 4: Process node B

Legend



Stack (top at right): M | J | G | F | C

Visited: A



DFS Traversal

Step 5: Discover L, E from B

Legend



Complete



Processing



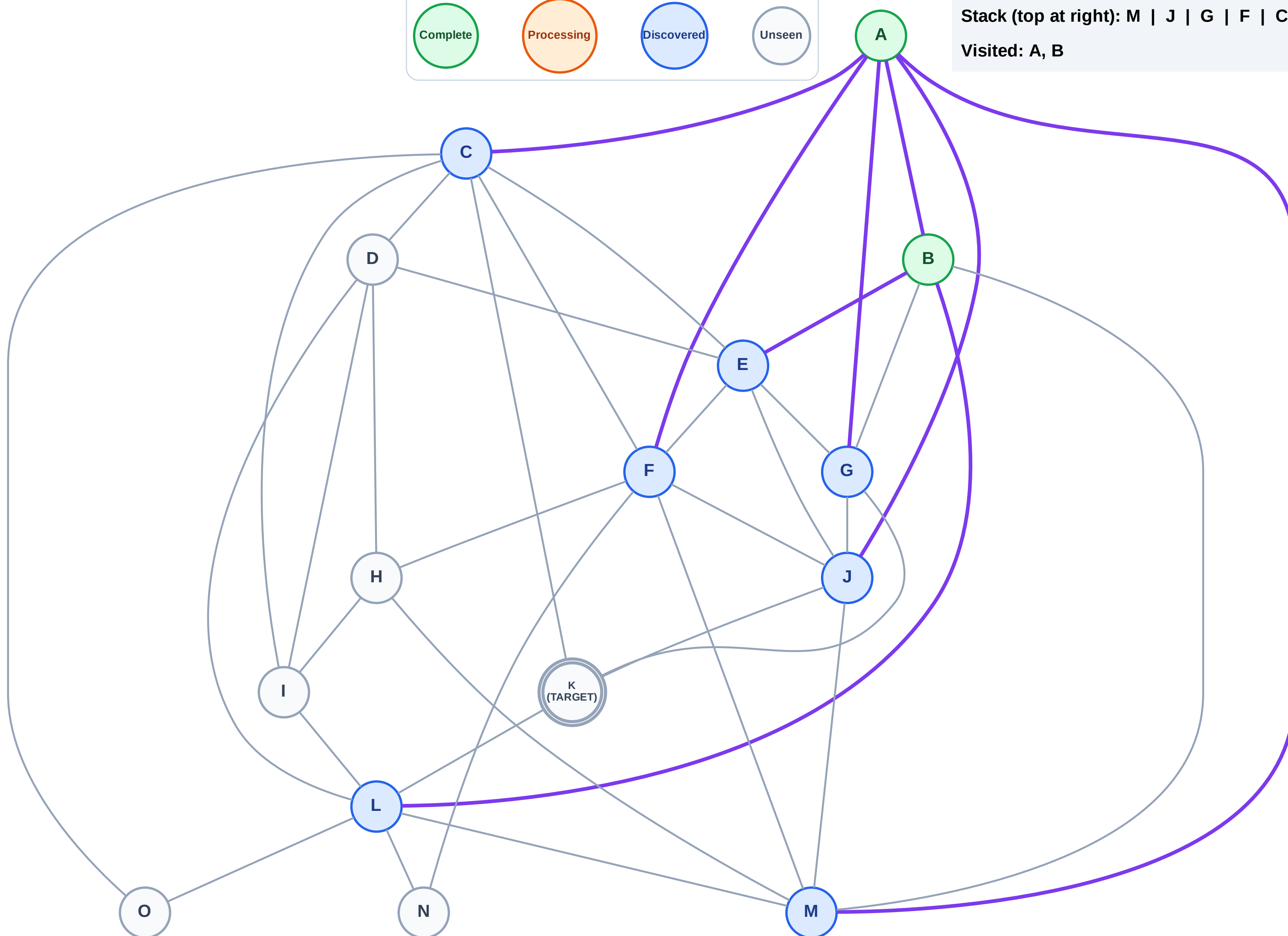
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | E

Visited: A, B



DFS Traversal

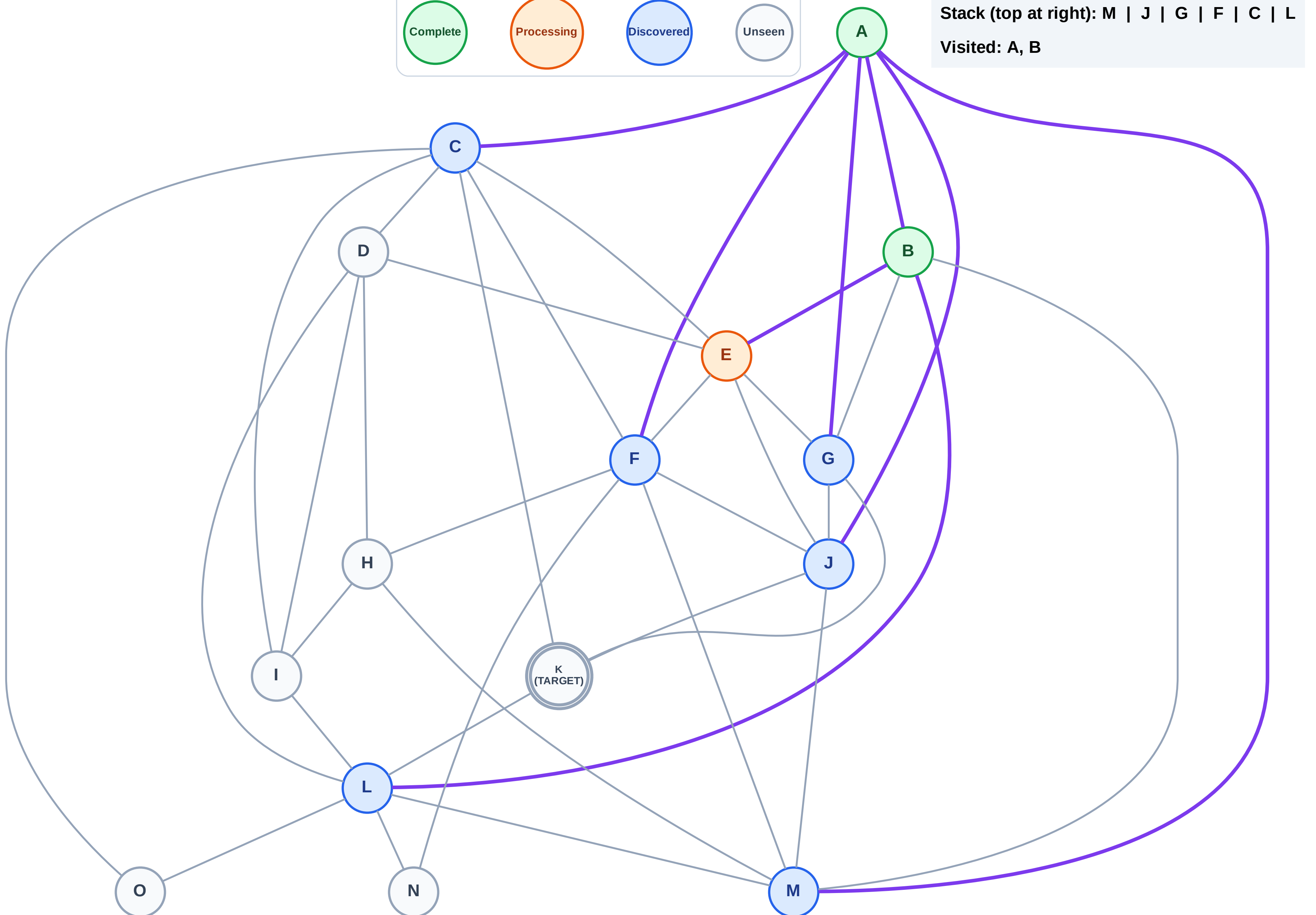
Step 6: Process node E

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B



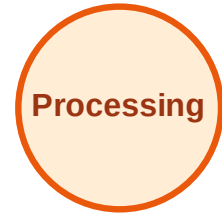
DFS Traversal

Step 7: Discover D from E

Legend



Complete



Processing



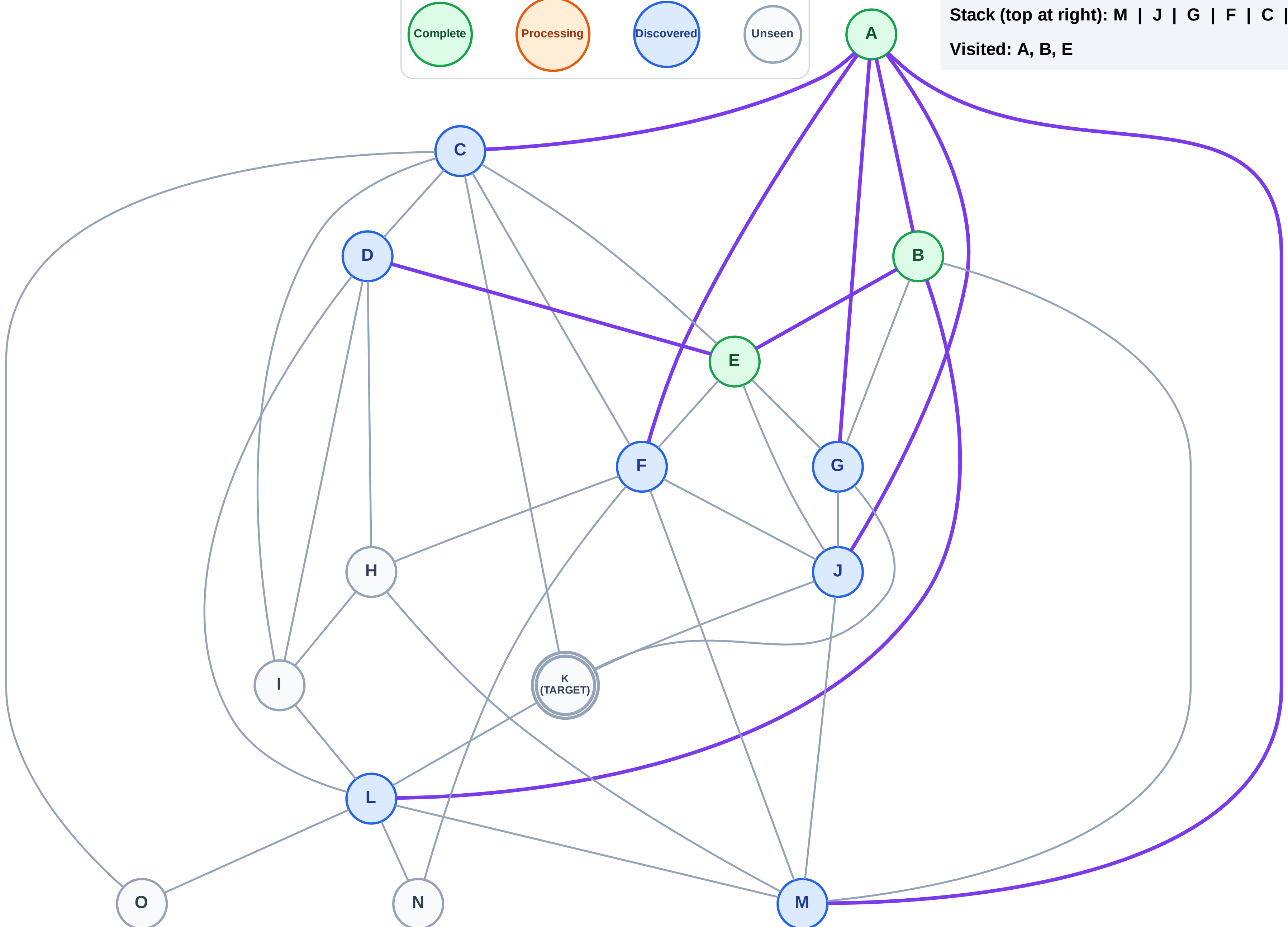
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | D

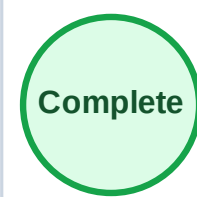
Visited: A, B, E



DFS Traversal

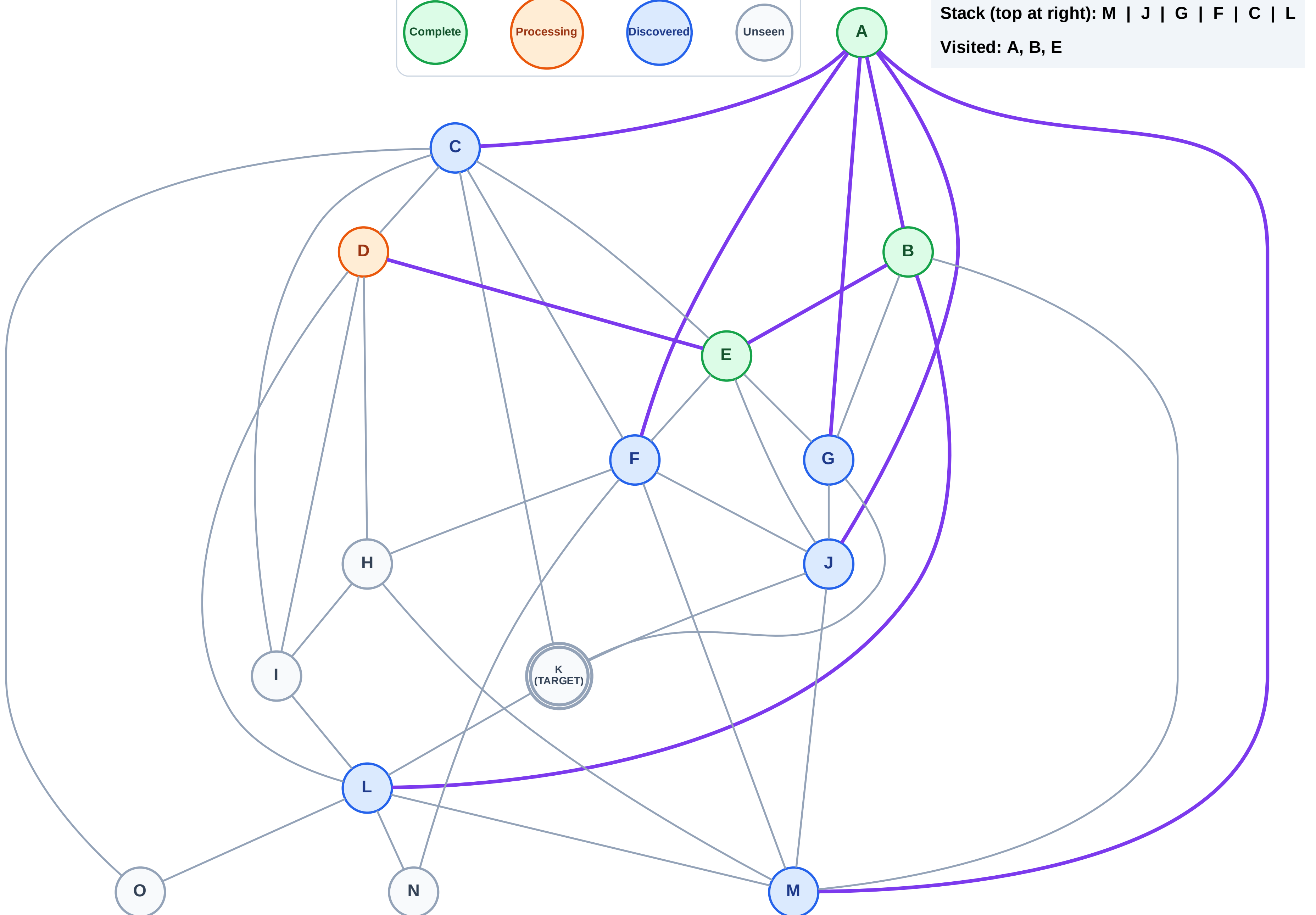
Step 8: Process node D

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B, E



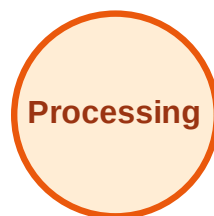
DFS Traversal

Step 9: Discover I, H from D

Legend



Complete



Processing



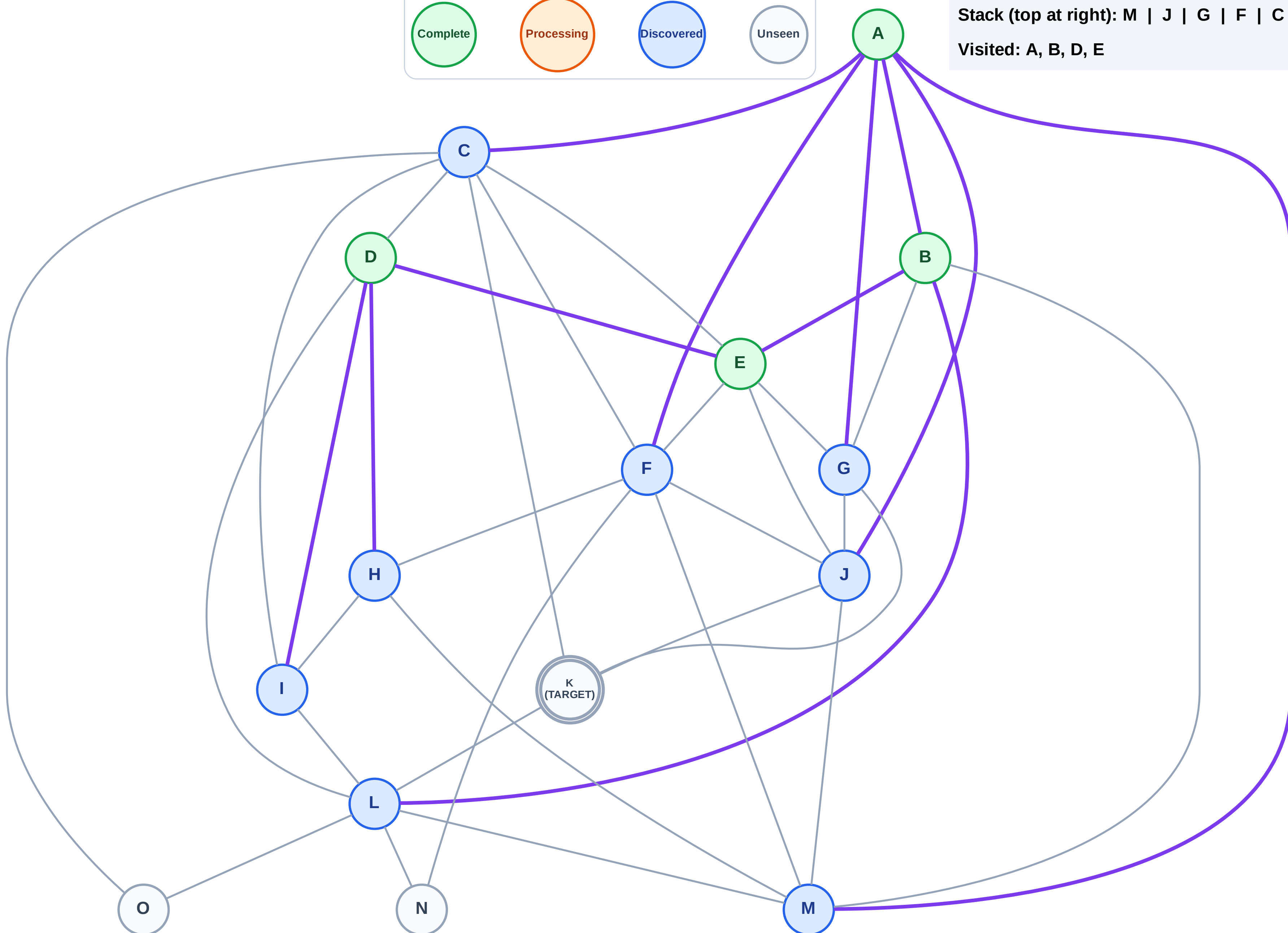
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I | H

Visited: A, B, D, E



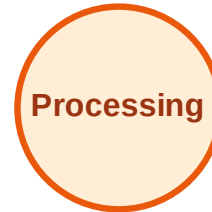
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



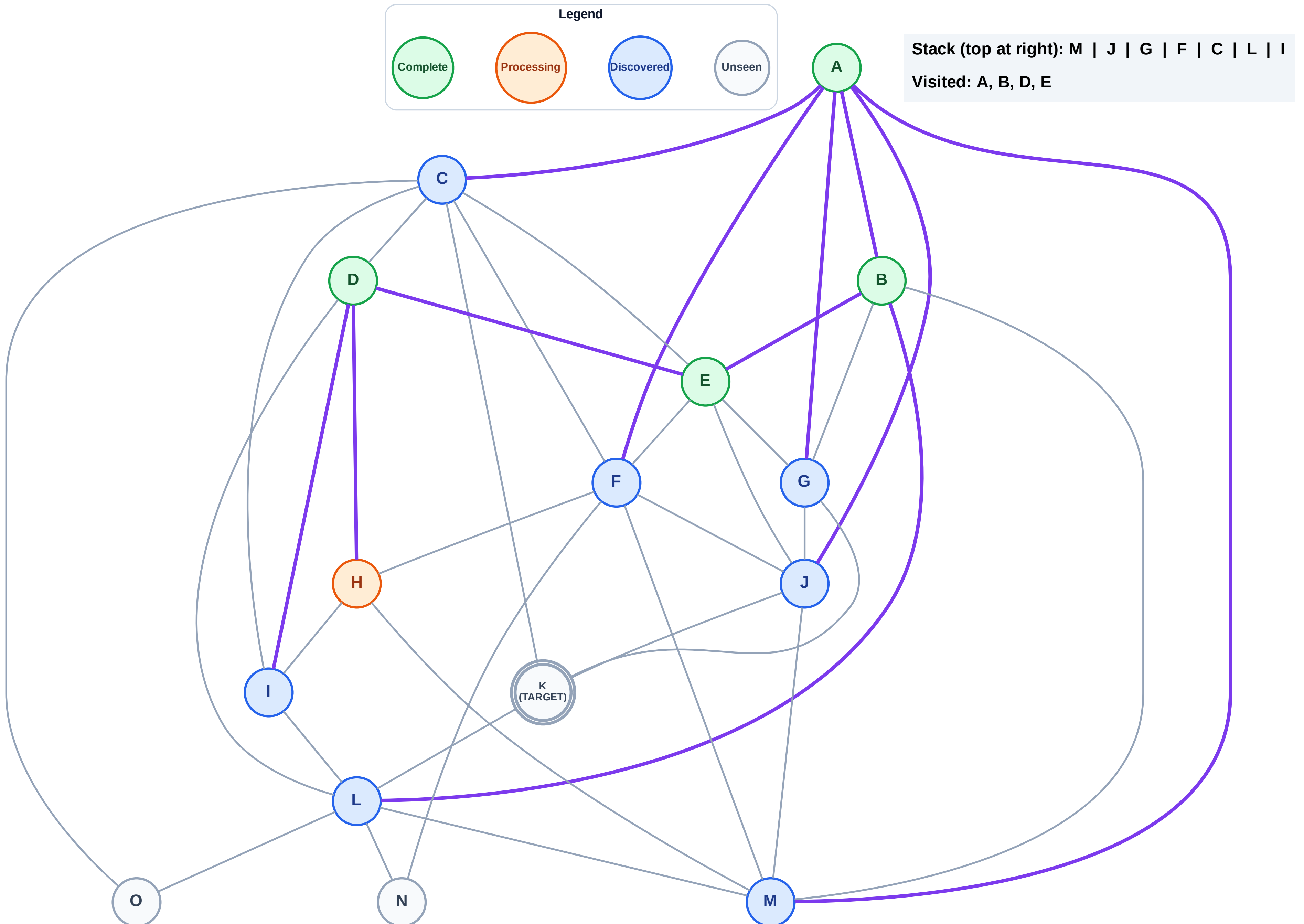
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I

Visited: A, B, D, E



DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



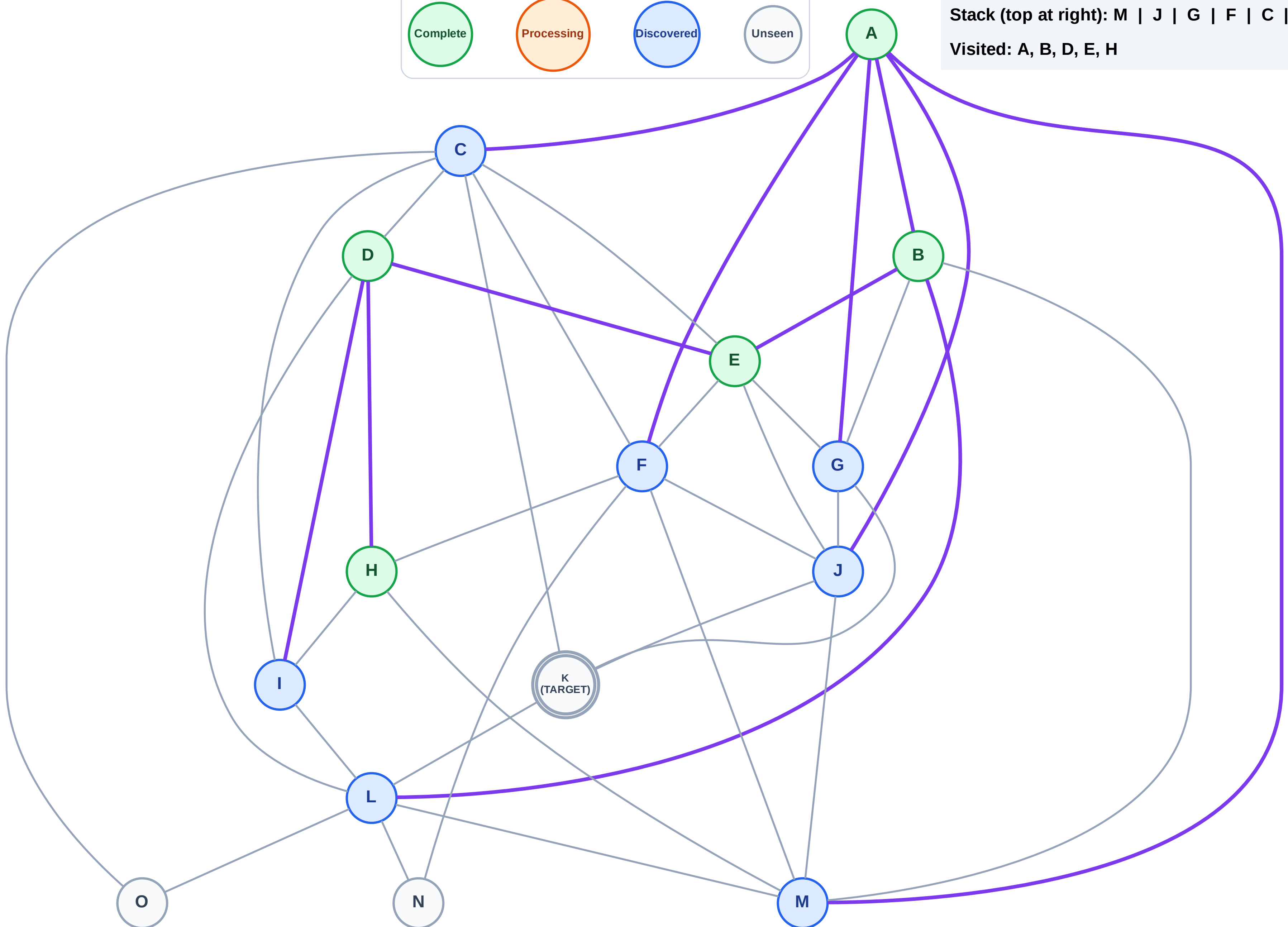
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I

Visited: A, B, D, E, H



DFS Traversal

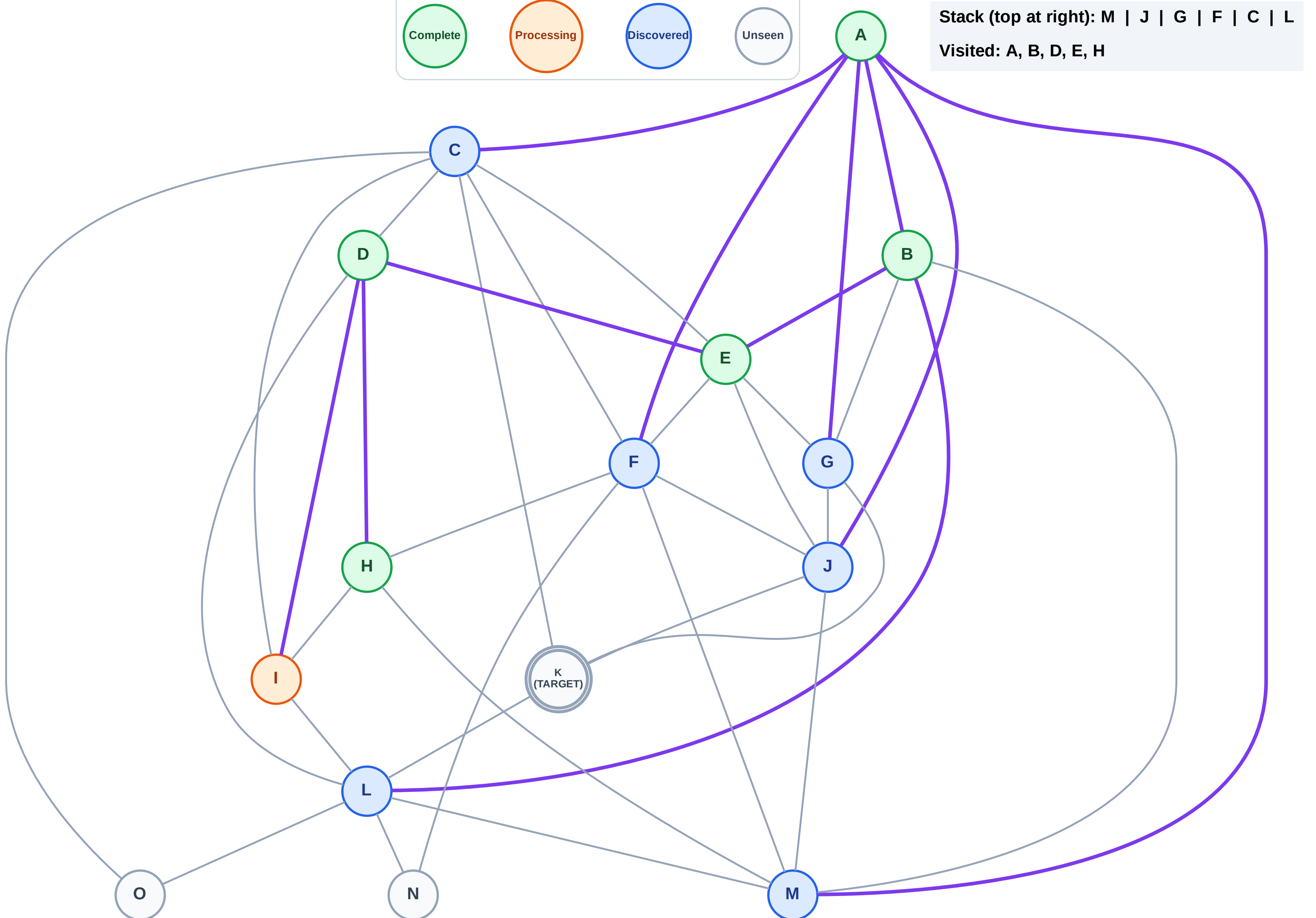
Step 12: Process node I

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B, D, E, H



DFS Traversal

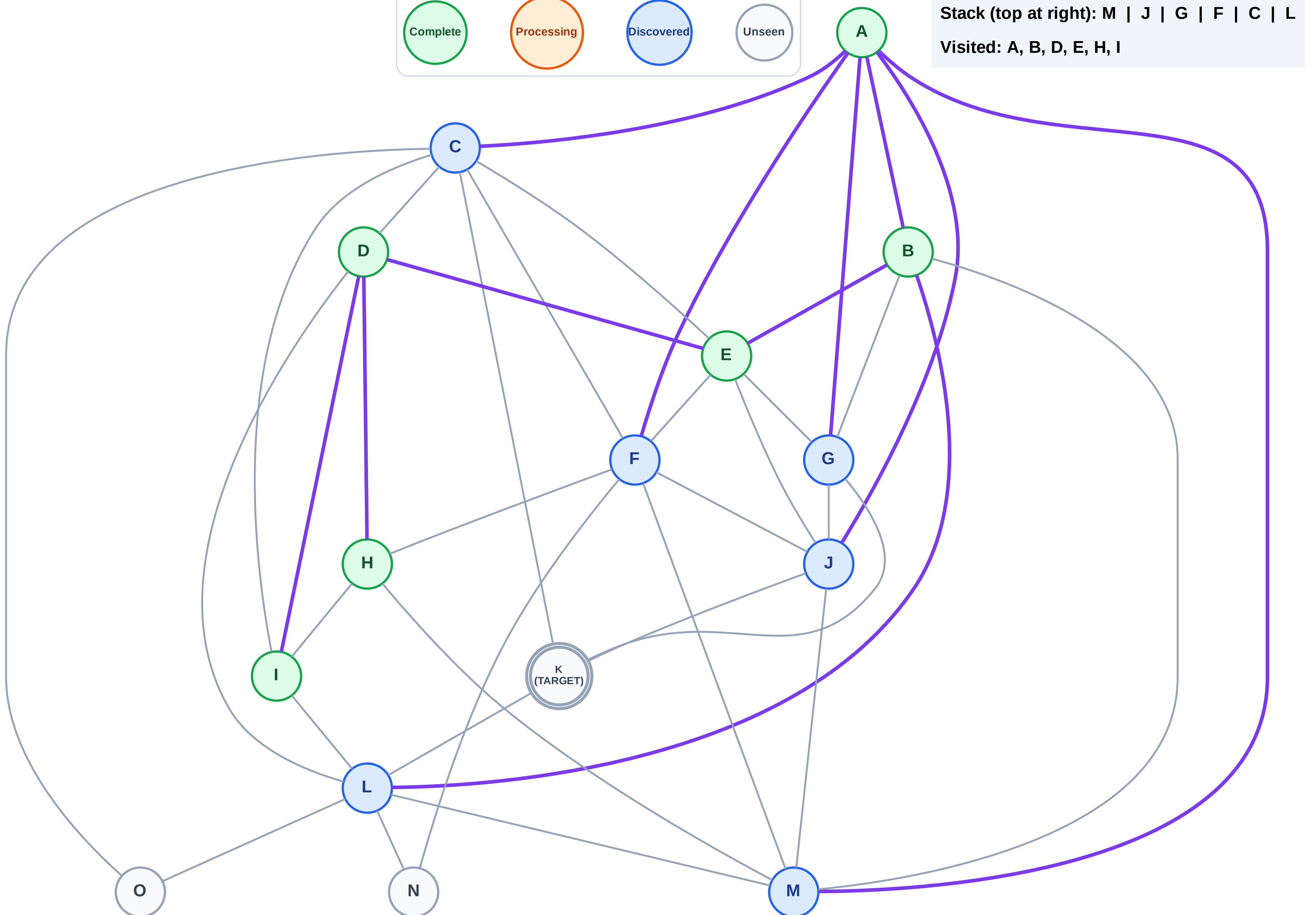
Step 13: No new neighbors from I

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B, D, E, H, I



DFS Traversal

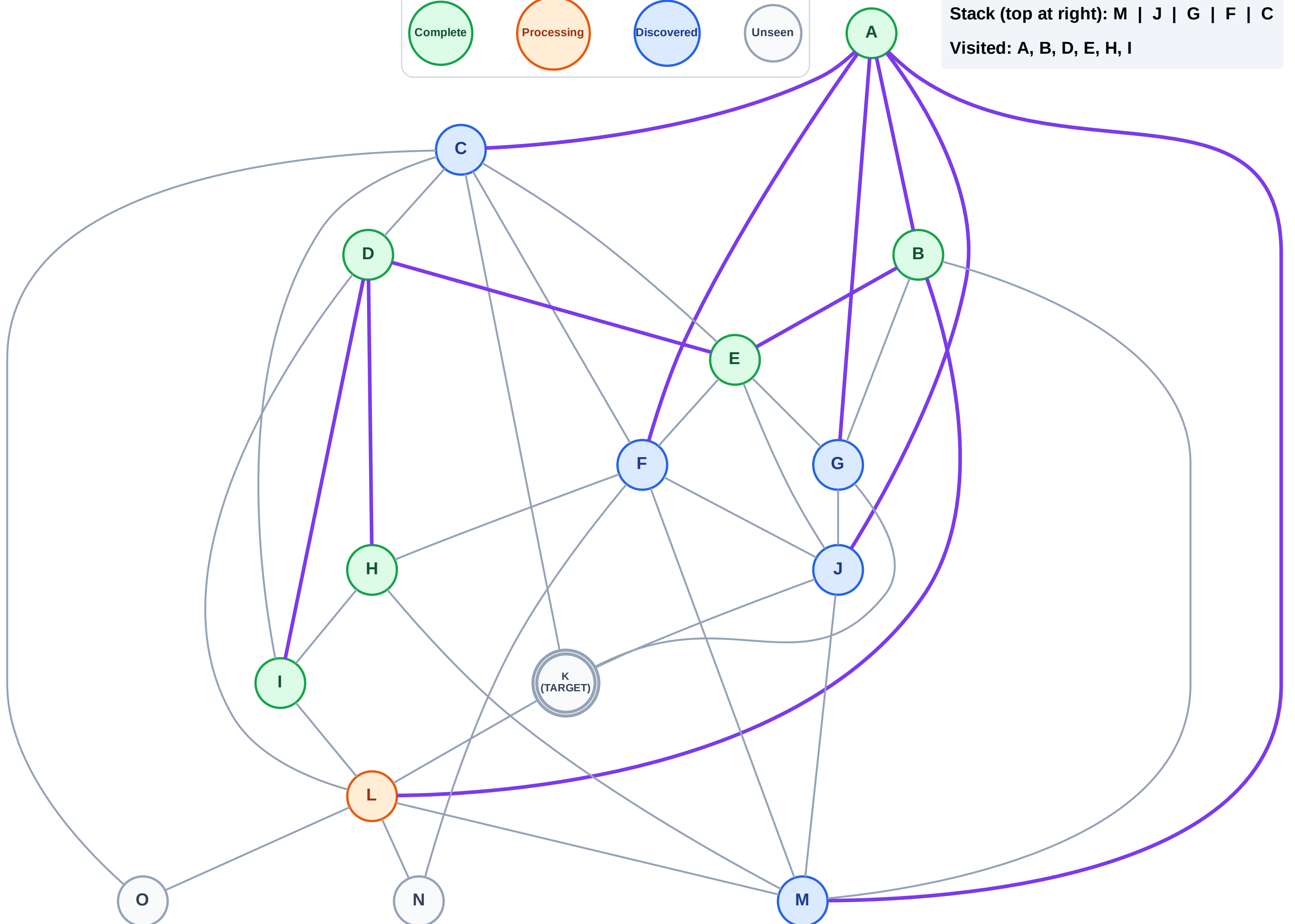
Step 14: Process node L

Legend



Stack (top at right): M | J | G | F | C

Visited: A, B, D, E, H, I



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → L → K

Legend



Stack (top at right): M | J | G | F | C | O | N

Visited: A, B, D, E, H, I, K, L

